F78AB - Actuarial and Financial Mathematics B
Tutorial 7 – Duration, Convexity and Immunisation

Question 1 Assume a constant force of interest of 6% p.a. and time is measured in years.

1. Calculate the Macaulay duration of the following sets of cashflows.

Set A: £300 p.a. paid continuously at a constant rate from $t = 0$ to $t = 10$.

Set B: £300 p.a. paid continuously at a constant rate from $t = 0$ to $t = 10$ and a discrete payment of £400 at $t = 3$.

Set C: a continuous payment paid from $t = 0$ to $t = 10$. The initial rate of payment of £300 p.a, and increases continuously thereafter at the rate 5% p.a. effective. Thus the rate of payment is $\rho(t) = 300(1.05)^t$, for $t \in [0, 10]$.

2. Using the calculated duration of Set A, explain why Set B has a lower duration than Set A. Does this depend on the chosen force of interest?

Question 2 An insurer has a liability to pay £150 million 8 years from now. The current market force of interest is 6% p.a. The insurer has available for investment a zero-coupon bond (ZCB) that matures in 5 years time and a ZCB that matures in 15 years' time.

1. Calculate the net present value, duration and convexity of the liabilities.

2. Calculate the nominal amounts of the ZCBs that need to be bought to ensure that the insurer is immunized against small changes in the force of interest.

3. By discounting the nominal amounts at the market force of interest, calculate the monetary amounts to be invested in each ZCB, i.e. how much money the insurer needs to spend now in order to buy the required nominal amounts of the ZCBs.

4. Verify that the convexity condition has been satisfied.

Question 3 Assuming that $NPV_A > 0$, prove that the three conditions in Redington's theorem are equivalent to:

- $NPV_A = NPV_L$,
- $NPV'_A = NPV'_L$, and
- $NPV''_A > NPV''_L$.

Question 4 An insurer has a liability to pay £10 million p.a. at the end of each of the next 15 years, i.e. at times $t = 1, \ldots, 15$. The current market force of interest is 4% p.a.

The insurer has available for investment a zero-coupon bond (ZCB) that matures in 2 years' time and a ZCB that matures in 20 years' time.

1. Using the conditions in Question 3, calculate the monetary amounts that need to be invested in the ZCBs to ensure that the insurer is immunized against small changes in the force of interest.

2. By accumulating the monetary amounts at the market force of interest, calculate the nominal amounts to be bought of each ZCB.

3. Verify that the convexity condition $NPV_A'' > NPV_L''$ has been satisfied. Note: you are advised to use a spreadsheet to calculate NPV_L''.

Question 5 An insurer has a liability to pay £5 million p.a. continuously at a constant rate for the next 12 years. The current market force of interest is 4% p.a. The insurer has available for investment a zero-coupon bond (ZCB) that matures in 15 years' time and a bank account, with which the insurer can borrow or deposit money at the market interest rate.

1. Explain why the duration of a sum of money deposited in the bank account is zero years.

2. Using the conditions in Question 3, calculate the monetary amounts that need to be invested in the 15-year ZCB and the bank account to ensure that the insurer is immunized against small changes in the force of interest.

3. By accumulating the monetary amount at the market force of interest, calculate the nominal amount to be bought of the 15-year ZCB.

4. Verify that the convexity condition has been satisfied. Hint: to calculate NPV_L'', apply integration-by-parts once and then note that $(\bar{I}\bar{a})_{\overline{n}|} = \int_0^n tv^t dt$.

Question 6 An investor has to pay a lump sum of £20 000 at the end of fifteen years from now and an annuity certain of £5 000 p.a. payable half-yearly in advance for twenty-five years, starting in ten years' time. The investor currently holds an amount of cash equal to the present value of these two liabilities calculated at the market rate of interest. The investor wishes to immunize her fund against small movements in the rate of interest by investing the cash in two ZCBs, Bond X and Bond Y. The investor has decided to invest an amount in Bond X sufficient to provide a capital sum of £25 000 when Bond X is redeemed in ten years' time. The remainder of the cash is invested in Bond Y.

In order to immunize her holdings at a market rate of interest of 7% p.a. effective:

1. Calculate the amount of money invested in Bond Y, and

2. Determine the term needed for Bond Y and the redemption amount payable on the maturity date.

3. Check if the investor is immunised by checking if the convexity condition $NPV_A'' > NPV_L''$ holds. Note: you are advised to use a spreadsheet to calculate NPV_L''. Can you explain your answer? What do you advise the investor to do?

Question 7 An investor borrows £100 000 to invest in a new company. Loans from the bank are subject to a force of interest of 8% p.a. Deposits in the bank earn interest at the rate of 5% p.a. effective.

The investment will produce the following cashflows only: Nothing during the first year, followed by a continuous payment stream of £15 000 per annum from time 1 to time 15, and finally a lump sum of £10 000 at time 15.

1. Calculate the discounted payback period on the loan.

2. Calculate the accumulated profit at time 15.

Numerical solutions: Q1.1. $\tau_A = 4.503$ years, $\tau_B = 4.309$ years, $\tau_C = 4.907$ years; Q2.1. $\text{NPV}_L =$ £92.818m, $\tau_L = 8$ years, $\text{conv}_L = 64$ years^2; Q2.2. £87.704m of 5-year ZCB, £68.489m of 15-year ZCB; Q2.3. £64.973m of 5-year ZCB, £27.846m of 15-year ZCB; Q4.1. £78.263m of 2-year ZCB, £32.293m of 20-year ZCB; Q4.2 £84.781m of 2-year ZCB, £71.869m of 20-year ZCB; Q5.2. £17.542m of 15-year ZCB, £30.110m in bank account; Q5.3. £31.964m; Q6.1. £25 706.99; Q6.2. Term 22.182 years, redemption proceeds £115 303.42; Q7.1. 11.777 years; Q7.2. £62 353.47.

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Solutions to Tutorial 7 – Duration, Convexity and Immunisation

Solution 1

1. As $\delta = 0.06$,

   Set A
   $$\frac{\int_0^{10} t \times 300 v^t dt}{\int_0^{10} 300 v^t dt} = \frac{(\bar{I}\bar{a})_{\overline{10}|}}{\bar{a}_{\overline{10}|}} = \frac{33.86150}{7.51981} = 4.503 \text{ years.}$$

   Set B
   $$\frac{3 \times 400v^3 + \int_0^{10} t \times 300 v^t dt}{400v^3 + \int_0^{10} 300 v^t dt} = \frac{1200v^3 + 300(\bar{I}\bar{a})_{\overline{10}|}}{400v^3 + 300\bar{a}_{\overline{10}|}} = \frac{1200v^3 + 300(33.86150)}{400v^3 + 300(7.51981)}$$
   $$= 4.309 \text{ years.}$$

   Set C
   $$\frac{\int_0^{10} t \times \rho(t) v^t dt}{\int_0^{10} \rho(t) v^t dt} = \frac{300 \int_0^{10} t \times (1.05)^t v^t dt}{300 \int_0^{10} (1.05)^t v^t dt}.$$

   Set $\hat{v} = 1.05v = e^{-(0.06 - \ln(1.05))}$, and define the equivalent force of interest $\hat{\delta} = 0.06 - \ln(1.05)$. Then the duration is
   $$\frac{\int_0^{10} t \times \hat{v}^t dt}{\int_0^{10} \hat{v}^t dt} = \frac{(\bar{I}\bar{a})^{\hat{\delta}}_{\overline{10}|}}{\bar{a}^{\hat{\delta}}_{\overline{10}|}} = \frac{46.41588}{9.45988} = 4.907 \text{ years.}$$

2. Set B is a combination of Set A and a discrete cashflow of £400. Since the duration of Set A is 4.503 years and the duration of the single discrete cashflow is 3 years, the duration of Set B is strictly between 3 and 4.503 years. Therefore, it is less than the duration of Set A.

   However, this depends on the force of interest. If a suitably high force of interest is used, the duration of Set A will fall below 3 years and the duration of Set B will be above the duration of Set A.

Solution 2 Work in £million.

1. $NPV_L = 150v^8 = 92.818$, $\tau_L = 8$ years and $\text{conv}_L = 64$ years^2.

2. Let u_5 be the nominal amount purchased of the 5-year ZCB, and let u_{15} be the nominal amount purchased of the 15-year ZCB. Then
   $$NPV_A = u_5 v^5 + u_{15} v^{15}$$
   and
   $$\tau_A = \frac{5 \times u_5 v^5 + 15 \times u_{15} v^{15}}{NPV_A}.$$
   The first two conditions of Redington's theorem give:
   $$NPV_A = NPV_L \qquad \Rightarrow \qquad u_5 v^5 + u_{15} v^{15} = 92.818 \tag{1}$$

and

$$\tau_A = \tau_L \qquad \Rightarrow \qquad \frac{5u_5v^5 + 15u_{15}v^{15}}{NPV_A} = 8.$$

As $NPV_A = NPV_L = 92.818$, the last equation becomes

$$5u_5v^5 + 15u_{15}v^{15} = 742.544. \tag{2}$$

Now we have two simultaneous linear equations (1) and (2) in two unknowns, viz u_5 and u_{15}. Solve to find $u_5 = 87.704$ and $u_{15} = 68.489$. Therefore, the insurer should purchase £87.704 million nominal of the 5-year ZCB and £68.489 million nominal of the 15-year ZCB.

3. To purchase £87.704 million nominal of the 5-year ZCB, the insurer pays £$87.704v^5$ million, i.e. £64.973 million today.

   To purchase £68.489 million nominal of the 15-year ZCB, the insurer pays £$68.489v^{15}$ million, i.e. £27.846 million today.

4. The convexity condition is $\text{conv}_A > \text{conv}_L$. As $\text{conv}_L = 64$ and

$$\text{conv}_A = \frac{5^2 \times 64.973 + 15^2 \times 27.846}{92.818} = 85 > 64,$$

   the convexity condition is satisfied.

Solution 3

- Follows directly.
- Since the NPV condition holds, multiplying both sides of $\tau_A = \tau_L$ by $-NPV_A$ gives the second condition $NPV'_A = NPV'_L$.
- Since the NPV condition holds and $NPV_A > 0$, multiplying both sides of $\text{conv}_A > \text{conv}_L$ by NPV_A gives the third condition $NPV''_A > NPV''_L$.

Solution 4 Work in £million. We use the equivalent conditions in question 3.

1. For the liabilities, we calculate

$$NPV_L = 10a_{\overline{15}|} = 10 \times 11.05562 = 110.556,$$

$$NPV'_L = -10(Ia)_{\overline{15}|} = -10 \times 80.23942 = -802.394,$$

   Let X_2 be the monetary amount purchased of the 2-year ZCB, and let X_{20} be the monetary amount purchased of the 20-year ZCB. Then

$$NPV_A = X_2 + X_{20}$$

   and

$$NPV'_A = -2 \times X_2 - 20 \times X_{20}.$$

   The first two conditions of Redington's theorem give:

$$NPV_A = NPV_L \qquad \Rightarrow \qquad X_2 + X_{20} = 110.556 \tag{3}$$

   and

$$NPV'_A = NPV'_L \qquad \Rightarrow \qquad 2X_2 + 20X_{20} = 802.394. \tag{4}$$

   Now we have two simultaneous linear equations (3) and (4) in two unknowns, viz X_2 and X_{20}. Solve to find $X_2 = 78.263$ and $X_{20} = 32.293$. Therefore, the insurer should purchase £78.263 million of the 2-year ZCB and £32.293 million of the 20-year ZCB.

2. Spending £78.263 million today will buy £$78.263e^{2\delta}$ million nominal of the 2-year ZCB, i.e. £84.781 million nominal.

   Spending £32.293 million today will buy £$32.293e^{20\delta}$ million nominal of the 20-year ZCB, i.e. £71.869 million nominal.

3. The convexity condition is $NPV''_A > NPV''_L$. Using a spreadsheet,

$$NPV''_L = \sum_{n=1}^{15} n^2 \times 10v^n = 7850.5$$

   and, without the need for a spreadsheet,

$$NPV''_A = 2^2 \times 78.263 + 20^2 \times 32.293 = 13230.3.$$

   Hence $NPV''_A > NPV''_L$ and convexity condition is satisfied.

   Note: if this was an exam question, you would be given the information $\sum_{n=1}^{15} n^2 \times 10v^n = 7850.5$.

Solution 5 Work in £million. We use the equivalent conditions in Question 3.

1. We can instantly take money out of the bank, i.e. we don't have to wait for a cashflow, so the duration is 0 years.

2. For the liabilities, we calculate

$$NPV_L = 5\bar{a}_{\overline{12}|} = 5 \times 9.53042 = 47.652,$$

$$NPV'_L = -5(\bar{I}\bar{a})_{\overline{12}|} = -5 \times 52.62536 = -263.127$$

   Let X_0 be the monetary amount deposited in the bank, and let X_{15} be the monetary amount purchased of the 15-year ZCB. Then

$$NPV_A = X_0 + X_{15}$$

   and

$$NPV'_A = -0 \times X_0 - 15 \times X_{15}.$$

   The first two conditions of Redington's theorem give:

$$NPV_A = NPV_L \qquad \Rightarrow \qquad X_0 + X_{15} = 47.652 \tag{5}$$

   and

$$NPV'_A = NPV'_L \qquad \Rightarrow \qquad 15X_{15} = 263.127 \qquad \Rightarrow \qquad X_{15} = 17.542. \tag{6}$$

   Substituting for X_2 in equation (5) gives $X_0 = 30.110$. Therefore, the insurer should purchase £17.542 million of the 15-year ZCB and deposit £30.110 million in the bank account.

3. Spending £17.542 million today will buy £$17.542e^{15\delta}$ million nominal of the 15-year ZCB, i.e. £31.964 million nominal.

4. The convexity condition is $NPV_A'' > NPV_L''$. Using integration-by-parts,

$$NPV_L'' = \int_0^{12} t^2 \times 5v^t dt = 2018.2$$

and

$$NPV_A'' = 0^2 \times 30.110 + 15^2 \times 17.542 = 3947.0.$$

Hence $NPV_A'' > NPV_L''$ and convexity condition is satisfied.

Note: you must be able to do integration-by-parts!

Solution 6

Let X denote the amount invested in Bond X and let Y denote the amount invested in Bond Y.

Let N denote the term to redemption of Bond Y, i.e. Bond Y is an N-year ZCB.

We are told that Bond X is redeemed in 10 years' time, i.e. Bond X is a 10-year ZCB.

1. First equate the NPVs of the assets and liabilities: $NPV_A = NPV_L$. As

$$NPV_L = 20\,000v^{15} + 5\,000_{10|}\ddot{a}^{(2)}_{\overline{25|}} = 7\,248.92 + 31\,166.80 = 38\,415.72,$$

then

$$NPV_A = X + Y = 38\,415.72.$$

As X is redeemed in 10 years' time for £25 000, we get

$$X = 25\,000v^{10} = 12\,708.73 \qquad \Rightarrow \qquad Y = 25\,706.99.$$

i.e £25 706.99 is invested in bond Y.

2. Next, the duration condition $NPV_A' = NPV_L'$. First,

$$-NPV_A' = 10X + NY = 127\,087.3 + 25\,706.99N. \tag{7}$$

We need to calculate NPV_L' in order to find N. There is a payment of 20 000 at time 15 and payments of 2 500 at times $10, 10.5, 11, \ldots, 34.5$. Hence

$$\begin{aligned} -NPV_L' &= 15 \times 20\,000v^{15} + \left(10 \times 2\,500v^{10} + 10.5 \times 2\,500v^{10.5} + \cdots + 34.5 \times 2\,500v^{34.5}\right) \\ &= 15 \times 20\,000v^{15} + 2\,500\left(10v^{10} + 10.5v^{10.5} + 11v^{11} + \cdots + 34.5v^{34.5}\right). \end{aligned}$$

Look for a geometric series in the term in brackets, i.e. something of the form $1 + r + r^2 + \cdots$.

$$\begin{aligned} -NPV_L' &= 15 \times 20\,000v^{15} + 2\,500\left((10+0)v^{10} + (10+0.5)v^{10.5} + \cdots + (10+24.5)v^{34.5}\right) \\ &= 15 \times 20\,000v^{15} + 2\,500\Big(10v^{10} + 10v^{10.5} + 10v^{11} + 10v^{11.5} + \cdots + 10v^{34.5} \\ &\qquad + 0.5v^{10.5} + 1v^{11} + 1.5v^{11.5} + \cdots + 24.5v^{34.5}\Big) \end{aligned}$$

As

$$v^{10} + v^{10.5} + v^{11} + \cdots + v^{34.5} = v^{10}\left(1 + v^{0.5} + v + \cdots + v^{24.5}\right) = v^{10}\ddot{a}^{j}_{\overline{50|}}$$

with $j = 1.07^{1/2} - 1$, and

$$\begin{aligned} 0.5v^{10.5} + 1v^{11} + 1.5v^{11.5} + \cdots + 24.5v^{34.5} &= 0.5v^{10}\left(v^{0.5} + 2v^1 + 3v^{1.5} + \cdots + 49v^{24.5}\right) \\ &= 0.5v^{10}\left((Ia)^{j}_{\overline{50|}} - 50v^{25}\right), \end{aligned}$$

Hence

$$\begin{aligned}
-NPV'_L &= 15 \times 20\,000v^{15} + 2\,500\left(10v^{10}\ddot{a}^{j}_{\overline{50|}} + 0.5v^{10}\left((Ia)^{j}_{\overline{50|}} - 50v^{25}\right)\right) \\
&= 15 \times 20\,000v^{15} + 2\,500v^{10}\left(10 \times 24.52389 + 0.5 \times \left(444.99582 - 50v^{25}\right)\right) \\
&= 108\,733.81 + 588\,580.26 \\
&= 697\,314.07.
\end{aligned}$$

Now we use the duration condition. From the last equation and equation (7),

$$NPV'_A = NPV'_L \qquad \Rightarrow \qquad 127\,087.3 + 25\,706.99N = 697\,314.07.$$

Solving, we find $N = 22.182$ years.

Thus, the investor must buy a 22.182-year ZCB. The redemption proceeds of this bond, i.e. Bond Y, are

$$Y(1.07)^N = 25\,706.99(1.07)^{22.182} = £115\,303.42.$$

3. Check if $NPV''_A > NPV''_L$.

$$NPV''_A = 10^2X + N^2Y = 10^2 \times 12\,708.73 + 22.182^2 \times 25\,706.99 = 13\,919\,769.3.$$

For NPV''_L, use a spreadsheet to calculate

$$\begin{aligned}
NPV''_L &= 15^2 \times 20\,000v^{15} \\
&\quad + \left(10^2 \times 2\,500v^{10} + 10.5^2 \times 2\,500v^{10.5} + \cdots + 34.5^2 \times 2\,500v^{34.5}\right) \\
&= 1\,631\,007.1 + 12\,529\,619.7 \\
&= 14\,160\,626.8
\end{aligned}$$

Therefore, at an effective rate of interest $i = 0.07$ p.a, the investor is not immunised as $NPV''_A < NPV''_L$.

The reason is that the liabilites are more "spread out" around the common duration

$$\tau = -\frac{NPV'_A}{NPV_A} = \frac{697\,314.07}{38\,415.72} = 18.152 \text{ years}$$

than are the assets. This means that the liabilities are more sensitive to changes in interest rates.

Mathematically, you can see this greater sensitivity by using a Taylor series expansion about the market force of interest δ_0 to get a second-order approximation for $NPV(\delta_0)$. Substitute into the approximation for the duration τ and convexity conv and you will see how the convexity affects the NPV as the interest rate changes.

The investor should seek alternative assets to invest in so that she is immunised.

Solution 7

1. Denote by $\delta_L = 0.08$ the force of interest on loans.

   The discounted payback period (DPP) is the first time when the discounted income from the investment exceeds the outstanding loan. Denote the DPP by T.

   We assume that the investor uses the income from the investment to pay off the loan first. This means that we use the the force of interest on loans to calculate the DPP T. We assume $T < 15$.

The income is received from $t = 1$ to $t = T$.

$$\begin{aligned} 100\,000 &\leq 15\,000 e^{-\delta_L} \bar{a}_{\overline{T-1}|}^{\delta_L} \\ \Rightarrow \frac{100\,000}{15\,000 e^{-\delta_L}} &\leq \frac{1 - e^{-\delta_L(T-1)}}{\delta_L} \\ \Rightarrow \frac{100\delta_L}{15 e^{-\delta_L}} &\leq 1 - e^{-\delta_L(T-1)} \\ \frac{8}{15 e^{-0.08}} &\leq 1 - e^{-0.08(T-1)} \\ \Rightarrow e^{-0.08(T-1)} &\leq 1 - \frac{8}{15 e^{-0.08}}. \end{aligned}$$

Take logs on both sides and don't forget to reverse the inequality when we divide by a negative number:

$$\begin{aligned} -0.08(T-1) &\leq \ln\left(1 - \frac{8}{15 e^{-0.08}}\right) \\ T - 1 &\geq -\frac{1}{0.08} \ln\left(1 - \frac{8}{15 e^{-0.08}}\right) \\ T &\geq 11.777. \end{aligned}$$

Continuous payments mean that we will have the discounted income less the outstanding loan exactly equal to zero at the DPP. This is unlike typical problems in discrete time where the bank account jumps from being strictly negative to being strictly positive.

Thus the DPP is 11.777 years.

2. Once the DPP time is reached, the loan has been paid off and all future income from the investment can be deposited at the bank at the interest rate 5% p.a. effective on bank deposits.

   The accumulated profit at $t = 15$ is

$$\begin{aligned} & 15\,000 \bar{s}_{\overline{15-11.777}|}^{5\%} + 10\,000 \\ &= 15\,000 \times 3.49023 + 10000 \\ &= 62\,353.47. \end{aligned}$$